SYSTEM VERIFICATION & QUALITY AUDIT

CareerTrack – Quality Assurance & Test Case Matrix

This document provides a comprehensive verification suite executed on the **CareerTrack Full-Stack Job Portal & Recruitment Management System**.

Test Execution Summary

30	30	0	100%
EXECUTED	PASSED	FAILED	QUALITY PASS

Detailed Test Cases & Results

Test Case ID	Feature / Module	Test Scenario	Input / Preconditions	Expected Result	Status
TC-PUB-01	Public Job Portal	Browse active published openings	Visitor visits `http://localhost:5173/`	Displays list of published jobs with badges & salary disclosure. Draft/expired excluded.	PASS
TC-PUB-02	Job Search & Filter	Full-text search and multi-filtering	Input `"React"`, filter Employment Type to `"Full-time"`	Only full-time React job positions are displayed; count updates dynamically.	PASS
TC-PUB-03	Job Details View	Detailed role breakdown page	Click on any job opening card	Navigates to `/job/:id` displaying full overview, responsibilities, and qualifications.	PASS
TC-APP-01	Candidate Application	Apply without registration or login	Provide valid candidate details and attach PDF résumé	Application accepted; HTTP 201 Created; returns reference ID (`CT-YYYY-XXXXX`).	PASS
TC-APP-02	Candidate Application	Duplicate application rejection	Submit duplicate application using previously applied email for same job	Request rejected with HTTP 409 Conflict with clear error message.	PASS
TC-APP-03	Candidate Application	Apply to closed or expired job opening	Attempt candidate application for closed job or past deadline	Rejected with HTTP 400 Bad Request; uploaded file cleaned up from server disk.	PASS
TC-APP-04	File Upload Handling	Résumé file extension validation	Attempt uploading `.exe` or `.txt` file instead of PDF/DOCX	Upload rejected with HTTP 400 Bad Request; file filter blocks disk storage.	PASS
TC-APP-05	File Upload Handling	Résumé file size limit enforcement	Attempt uploading résumé exceeding 5 MB (e.g., 8 MB file)	Multer triggers `LIMIT_FILE_SIZE`; HTTP 400 response returned with message.	PASS
TC-AUTH-01	Administrator Auth	Administrator authentication	Submit valid credentials (`admin@careertrack.com` / `AdminPassword123!`)	Returns JWT token and admin profile; navigates to `/admin/dashboard`.	PASS
TC-AUTH-02	Administrator Auth	Invalid password rejection	Submit incorrect administrator password	Returns HTTP 401 Unauthorized; generic error message prevents enumeration.	PASS
TC-AUTH-03	Security & Auth	Protect administrative routes	Unauthenticated client attempts `GET /api/jobs/admin/all`	Request rejected with HTTP 401 Unauthorized; access denied returned.	PASS

Test Case ID	Feature / Module	Test Scenario	Input / Preconditions	Expected Result	Status
TC-DASH-01	Admin Dashboard	Telemetry metrics calculation	Fetch summary `GET /api/jobs/admin/analytics/summary`	Accurate counts for total jobs, active jobs, closed jobs, and candidate pipeline.	PASS
TC-JOB-01	Job Management	Create new job posting	Submit complete job details through administrator modal	Job inserted into MongoDB with `createdBy` reference; HTTP 201 returned.	PASS
TC-JOB-02	Job Management	Toggle publication status	Change job status from `Published` to `Unpublished`	Job status updated in DB; immediately disappears from public listings.	PASS
TC-JOB-03	Job Management	Delete job posting with cascade	Delete job opening that has candidate applications	Job posting and all associated candidate application records removed from DB.	PASS
TC-RECRUIT-01	Application Management	Update candidate recruitment status	Change candidate status from `Submitted` to `Shortlisted`	Status updated in database; badge updates instantly in table & drawer UI.	PASS
TC-RECRUIT-02	Application Management	Add internal recruitment note	Administrator enters evaluation note in review drawer	Note appended to application's `notes` array with author name & timestamp.	PASS
TC-EXP-01	Application Management	Export applications to CSV	Click "Export Candidate CSV" button	Server streams clean CSV file with headers, candidate data, and status.	PASS
TC-EMAIL-01	Gmail Notification	Candidate confirmation email on application submit	Valid candidate submits application for active job	Candidate receives branded HTML confirmation email with Ref ID and Job metadata.	PASS
TC-EMAIL-02	Gmail Notification	Admin new applicant alert email on application submit	Valid candidate submits application for active job	Admin Gmail receives HTML alert with candidate profile details & skills.	PASS
TC-EMAIL-03	Gmail Notification – Status	"Under Review" status change email to candidate	Admin changes candidate status to `Under Review`	Candidate receives HTML email confirming application is under active review.	PASS
TC-EMAIL-04	Gmail Notification – Status	"Shortlisted" status change email to candidate	Admin changes candidate status to `Shortlisted`	Candidate receives congratulatory HTML shortlisting notice with purple pill.	PASS
TC-EMAIL-05	Gmail Notification – Status	"Interview Scheduled" status change email to candidate	Admin changes candidate status to `Interview Scheduled`	Candidate receives interview invitation email with follow-up details.	PASS
TC-EMAIL-06	Gmail Notification – Status	"Selected" status change email to candidate	Admin changes candidate status to `Selected`	Candidate receives selection offer HTML email referencing HR onboarding contact.	PASS
TC-EMAIL-07	Gmail Notification – Status	"Rejected" status change email to candidate	Admin changes candidate status to `Rejected`	Candidate receives respectful rejection HTML email maintaining positive tone.	PASS

Test Case ID	Feature / Module	Test Scenario	Input / Preconditions	Expected Result	Status
TC-EMAIL-08	Gmail Notification – Admin Direct	Admin sends custom direct email to candidate	Admin opens compose modal, fills body, and dispatches email	Custom email dispatched to candidate; action logged in internal recruiter notes thread.	PASS
TC-EMAIL-09	Gmail Notification – Admin Direct	Admin sends "Interview Scheduled" template email	Admin selects "Interview Scheduled" template in modal	Template populates interview text with candidate name, job title, and company.	PASS
TC-EMAIL-10	Gmail Notification – Contact Us	User submits Contact Us form → Admin receives inquiry	User submits Contact Us form on `/contact`	Admin Gmail receives formatted HTML inquiry email with message content.	PASS
TC-EMAIL-11	Gmail Notification – Contact Us	Contact Us form sends acknowledgement email to user	User submits Contact Us form on `/contact`	User receives branded CareerTrack acknowledgement email with message copy.	PASS
TC-EMAIL-12	Gmail Notification – SSL/TLS	Email delivery through corporate proxy / antivirus	Server runs behind local SSL-intercepting antivirus/proxy	`tls: { rejectUnauthorized: false }` bypasses chain error; email dispatches successfully.	PASS

Conclusion: All mandatory functional requirements, authentication gates, validation rules, duplicate prevention mechanisms, file upload constraints, recruitment workflows, and the complete **Gmail Notification System** have completed verification successfully with **zero defects**.